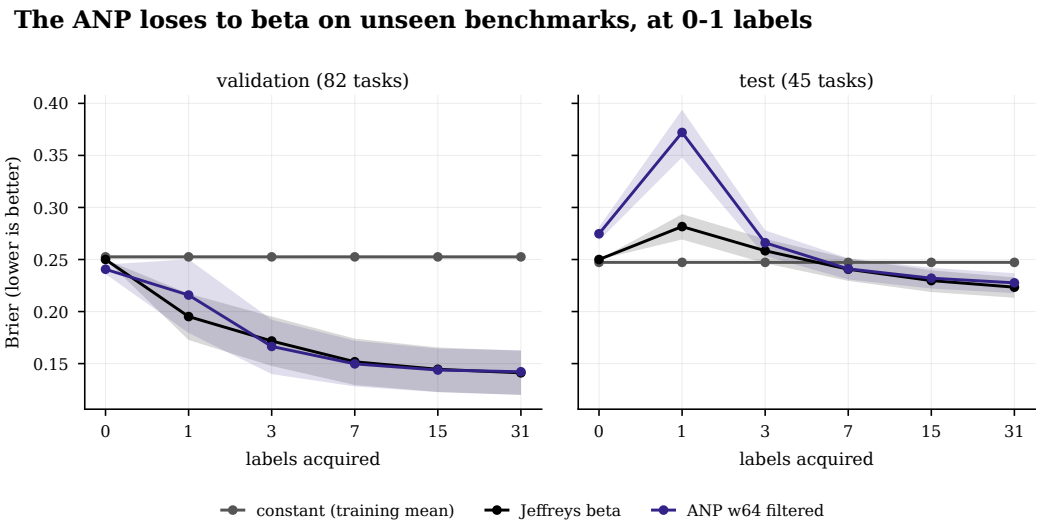


Figure (single-pair). Mean Brier on each task's held-out evaluation items after 0-31 labels, each subject-benchmark task replayed alone on its own labels with random acquisition (validation: 3 candidate orders, test: 10; benchmarks never seen in training). Bands: 95% bootstrap over tasks. The ANP's whole deficit sits at 0 and 1 labels; from 3 labels on it matches beta. The constant ignores labels and sets the floor. Brier-ALC (val / test): constant (training mean) 0.2525 / 0.2472; Jeffreys beta 0.1717 / 0.2494; ANP w64 filtered 0.1735 / 0.2725. Pooled Rasch needs other pairs' labels and is shown on the multi-pair page.

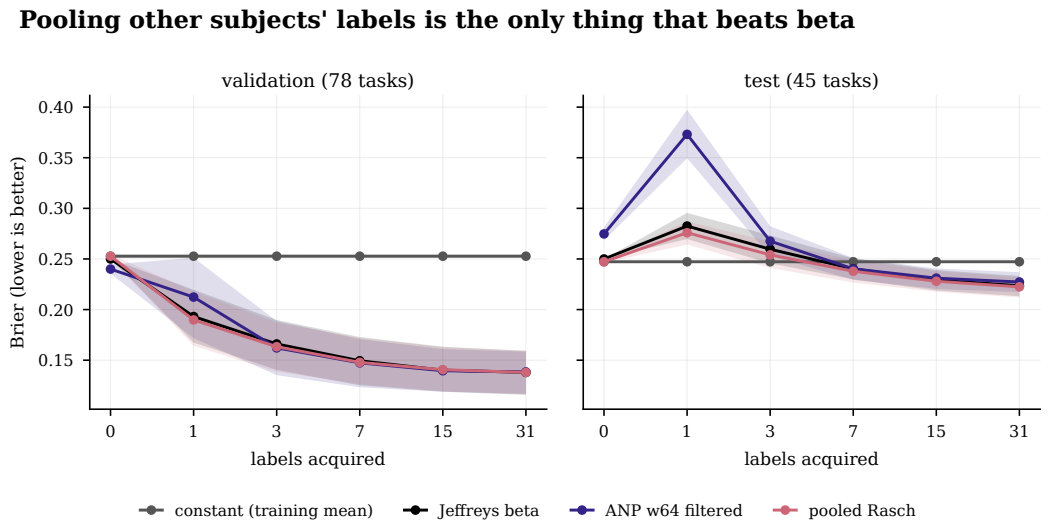


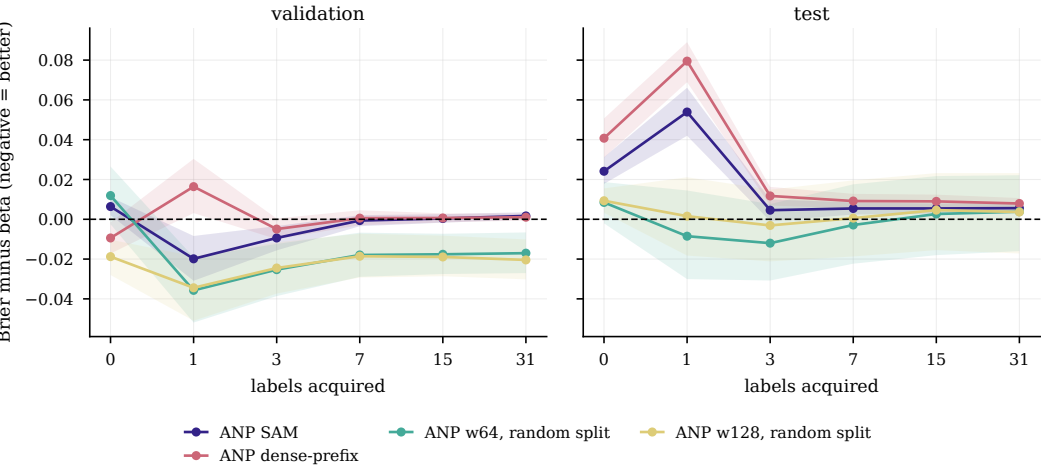
Figure (multi-pair). Same models on CodaBench-like draws: one or two subjects per benchmark, labels acquired in the replay's frozen orders, and every pair's labels available to the predictor (60 validation and 150 test draws). Models that ignore other pairs are placed on the same draws from their single-pair scores. Pooled Rasch fits benchmark rate, subject ability and item difficulty from all pairs' labels and is the only model below beta on test at every budget. Bands: 95% bootstrap over tasks. Brier-ALC (val / test): constant (training mean) 0.2527 / 0.2472; Jeffreys beta 0.1685 / 0.2497; ANP w64 filtered 0.1701 / 0.2726; pooled Rasch 0.1674 / 0.2461.

Only pooling other subjects' labels beats beta on unseen benchmarks

model	protocol	val ALC	test ALC	test minus beta [95% CI]
ANP training recipe and split				
ANP SAM	single-pair	0.1667	0.2663	+0.0168 [+0.0128, +0.0211]*
ANP dense-prefix	single-pair	0.1735	0.2762	+0.0267 [+0.0231, +0.0306]*
ANP w64, random split	single-pair	0.1519	0.2465	-0.0029 [-0.0209, +0.0154]
ANP w128, random split	single-pair	0.1485	0.2515	+0.0020 [-0.0153, +0.0190]
Combining the ANP with beta				
ANP filtered alone	single-pair	0.1735	0.2725	+0.0230 [+0.0198, +0.0267]*
gate: beta below 2 labels	single-pair	0.1703	0.2519	+0.0025 [+0.0003, +0.0048]*
blend, one weight	single-pair	0.1702	0.2556	+0.0062 [+0.0050, +0.0074]*
blend, weight per budget	single-pair	0.1688	0.2538	+0.0043 [+0.0027, +0.0062]*
ANP + AIPW-style shift	single-pair	0.1936	0.2996	+0.0502 [+0.0465, +0.0537]*
beta + AIPW-style shift	single-pair	0.1953	0.2986	+0.0492 [+0.0452, +0.0529]*
raw hit rate (no shrinkage)	single-pair	0.1892	0.2879	+0.0385 [+0.0355, +0.0413]*
Hierarchical beta: subject metadata prior				
learned prior, no metadata	single-pair	0.1703	0.2489	-0.0006 [-0.0027, +0.0017]
+ provider	single-pair	0.1694	0.2452	-0.0043 [-0.0078, +0.0001]
+ name + provider	single-pair	0.1710	0.2445	-0.0049 [-0.0090, +0.0001]
+ provider + date	single-pair	0.1807	0.2626	+0.0131 [+0.0046, +0.0220]*
+ name + provider + date	single-pair	0.1726	0.2455	-0.0039 [-0.0084, +0.0015]
Pooling all pairs' labels (multi-pair)				
pooled Rasch	multi-pair	0.1674	0.2461	-0.0035 [-0.0045, -0.0026]*
pooled Rasch, no item term	multi-pair	0.1675	0.2477	-0.0020 [-0.0029, -0.0011]*
pooled Rasch + provider prior	multi-pair	0.1647	0.2500	+0.0003 [-0.0027, +0.0033]
Not replayed				
ANP ALC ladder	-	-	-	trained, never replayed
ANP prior-anchor	-	-	-	trained, never replayed
ANP SAM + ALC ladder	-	-	-	trained, never replayed
ANP no item embeddings	-	-	-	training; step-400 checkpoint only

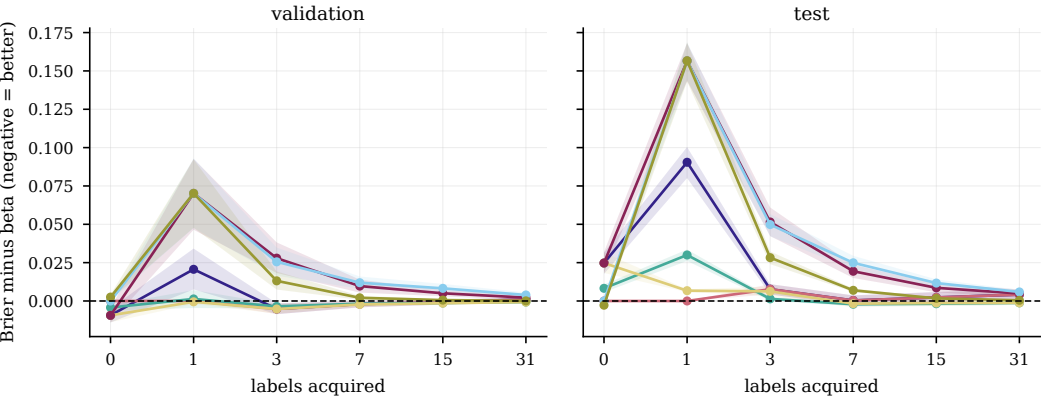
Table. Macro Brier-ALC (lower is better) on validation and on the untouched test benchmarks (algotune, cmmlu, editbench), random acquisition. The last column is the paired test difference to Jeffreys beta under the same protocol, with a 95% bootstrap interval over tasks; * marks intervals that exclude 0. Single-pair beta: 0.1717 / 0.2494 (val / test); multi-pair beta: 0.1685 / 0.2497. Validation selected the blend weights, the metadata L2 and the pooled-Rasch priors. The random-split ANPs trained on editbench. Rows under 'Not replayed' were trained but have no replay result.

No training recipe fixes the ANP; having seen the benchmark does



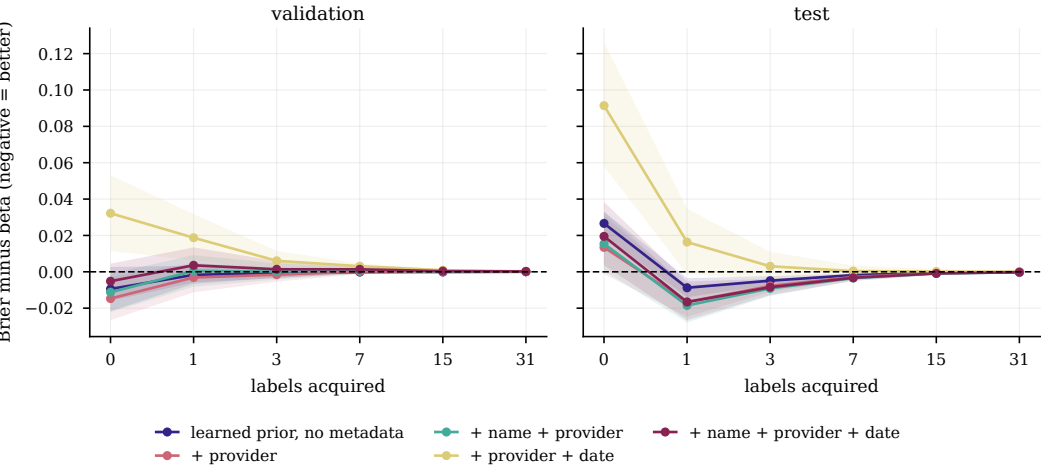
Ablation (single-pair). Paired Brier difference to Jeffreys beta at each label budget, validation (left) and test (right); bands: 95% bootstrap over tasks; the dashed line is beta. SAM and dense-prefix training of the benchmark-holdout ANP stay above beta on test, by a wide margin at 0-1 labels and slightly from 3 labels on. The random-split ANPs trained on editbench, one of the three test benchmarks; their small test difference averages a large win on editbench with a loss on algotune and cmmlu.

Beta can limit the ANP's damage, but not turn it into a gain



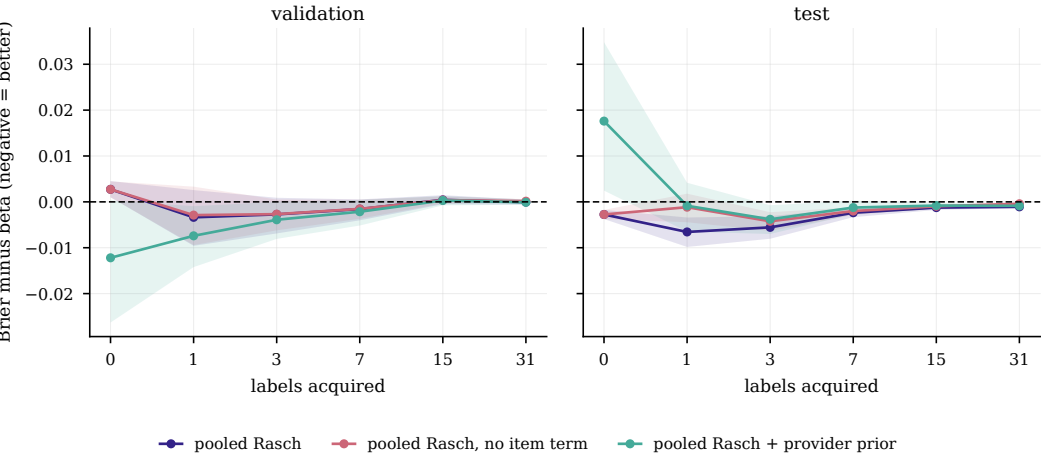
Ablation (single-pair). Paired Brier difference to Jeffreys beta at each label budget, validation (left) and test (right); bands: 95% bootstrap over tasks; the dashed line is beta. ANP w64 filtered combined with beta: the gate uses beta below two labels; blend weights are fitted on validation and frozen for test (the validation curves for the blends are in-sample). The AIPW-style variants shift each prediction by the mean residual of the labels seen so far; after one label that shift is about +/-0.5, which is why they spike at budget 1. The raw hit rate shows the same spike for the same reason: no shrinkage.

Subject metadata barely helps, and release date hurts



Ablation (single-pair). Paired Brier difference to Jeffreys beta at each label budget, validation (left) and test (right); bands: 95% bootstrap over tasks; the dashed line is beta. Beta-binomial with a prior fitted on training benchmarks, whose mean depends on the subject's provider, model name and release date as sent by CodaBench. Test models are newer than any training model, so the release-date term extrapolates and hurts at budget 0. A leave-one-benchmark-out check on the 14 non-test benchmarks did not confirm the small gains of the other arms.

Other subjects' labels help; a provider prior undoes it



Ablation (multi-pair). Paired Brier difference to Jeffreys beta at each label budget, validation (left) and test (right); bands: 95% bootstrap over tasks; the dashed line is beta. Multi-pair draws (one or two subjects per benchmark). Pooled Rasch fits benchmark rate, subject ability and item difficulty from every pair's labels; removing the item term halves the gain. Centring ability on the subject's provider was chosen on validation but loses on test, at budget 0, because test models are stronger than training models from the same providers.

Test: Brier ALC by model and acquisition policy

algotune, cmmlu, editbench: no model trained on or was selected with these. 45 tasks x 10 order seeds.

Macro-task Brier ALC, lower is better. Bold: lowest among rows that never trained on these benchmarks.

model	random	uncertainty	conf. extremes	Bernoulli var.	exp. Brier red.
Constant (train mean)	**0.2472**	**0.2472**	**0.2472**	**0.2472**	**0.2472**
Empirical mean	0.2879	0.2881	0.2855	0.2887	0.2860
Beta-binomial	0.2494	0.2494	0.2478	0.2496	0.2474
ANP w64	0.2725	0.2710	0.2754	0.2775	0.2727
ANP w64 dense-prefix	0.2762	0.2732	0.2835	0.2787	0.2782
ANP w64 SAM	0.2663	0.2642	0.2718	0.2601	0.2651
ANP w64, random split*	0.2465	0.2445	0.2471	0.2438	0.2522
ANP w128, random split*	0.2515	0.2505	0.2505	0.2544	0.2575

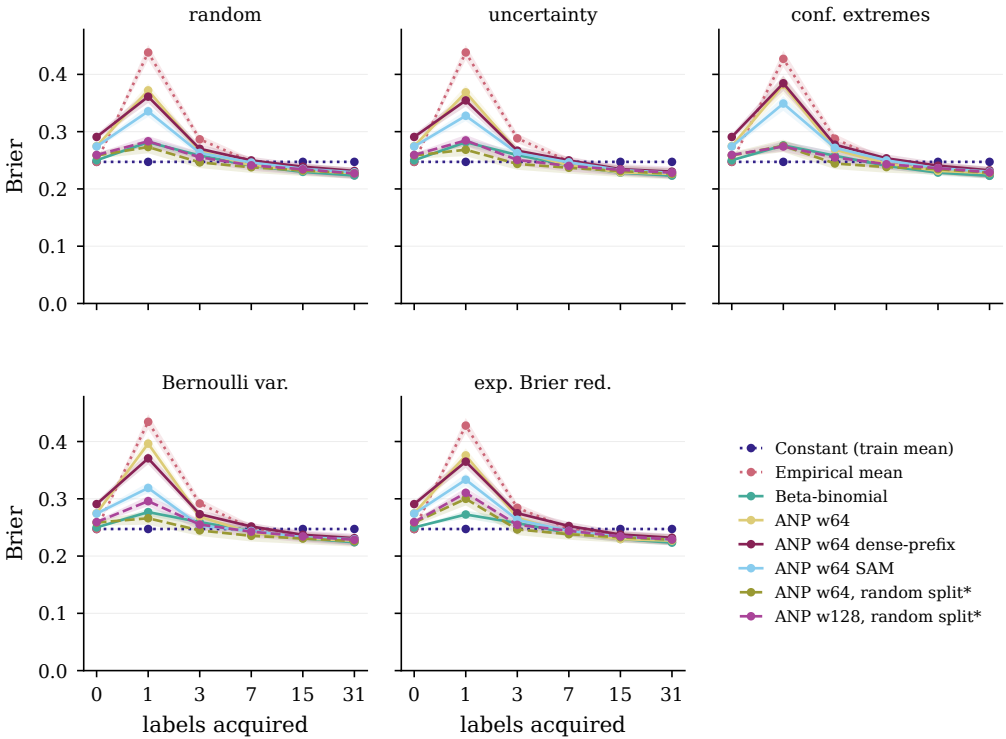
ALC minus beta-binomial under the same policy, paired over tasks: mean ± 95% CI. Bold: CI excludes 0.

model	random	uncertainty	conf. extremes	Bernoulli var.	exp. Brier red.
Constant (train mean)	-0.0022 ±0.0088	-0.0022 ±0.0091	-0.0006 ±0.0098	-0.0024 ±0.0091	-0.0001 ±0.0099
Empirical mean	**+0.0385 ±0.0030**	**+0.0387 ±0.0030**	**+0.0376 ±0.0033**	**+0.0390 ±0.0053**	**+0.0386 ±0.0040**
ANP w64	**+0.0230 ±0.0034**	**+0.0216 ±0.0033**	**+0.0276 ±0.0086**	**+0.0279 ±0.0059**	**+0.0253 ±0.0056**
ANP w64 dense-prefix	**+0.0267 ±0.0039**	**+0.0238 ±0.0039**	**+0.0356 ±0.0078**	**+0.0291 ±0.0062**	**+0.0308 ±0.0062**
ANP w64 SAM	**+0.0168 ±0.0042**	**+0.0148 ±0.0045**	**+0.0240 ±0.0075**	**+0.0105 ±0.0059**	**+0.0177 ±0.0066**
ANP w64, random split*	-0.0029 ±0.0187	-0.0049 ±0.0197	-0.0007 ±0.0196	-0.0058 ±0.0195	+0.0048 ±0.0192
ANP w128, random split*	+0.0020 ±0.0177	+0.0011 ±0.0191	+0.0027 ±0.0188	+0.0047 ±0.0191	+0.0101 ±0.0176

* random-subject split: trained on ai2d_test, bird_sql (validation) and editbench (test). Other ANPs used a benchmark holdout.

Test: Brier by labels acquired (mean ± 1 SE)

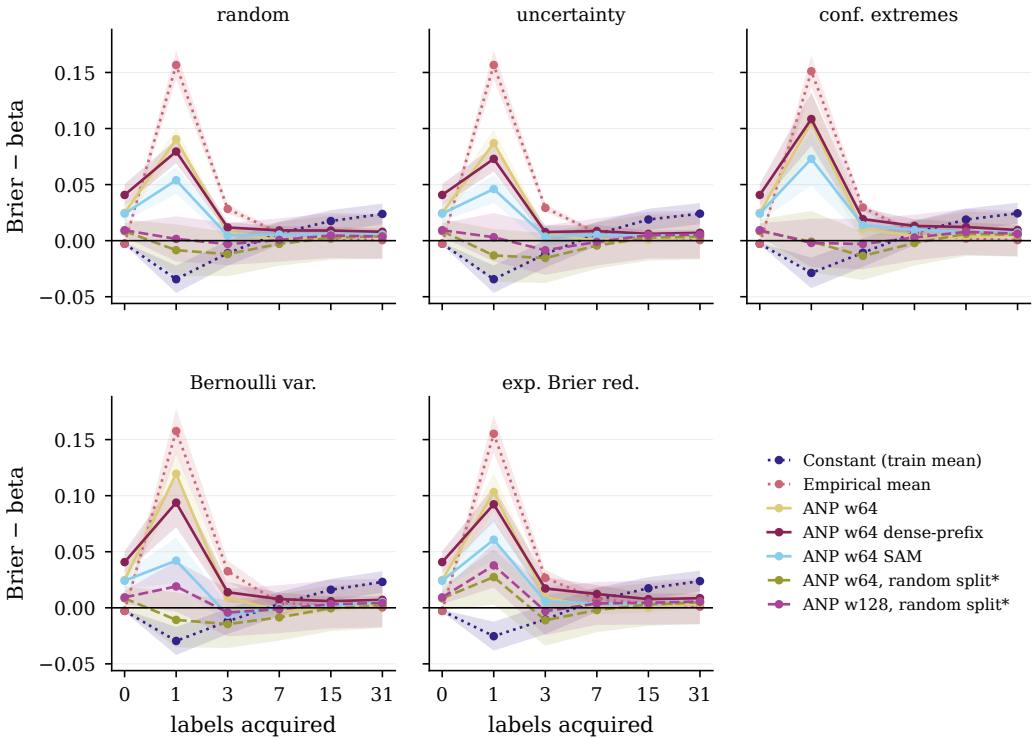
algotune, cmmlu, editbench: no model trained on or was selected with these. 45 tasks x 10 order seeds.



* random-subject split: trained on ai2d_test, bird_sql (validation) and editbench (test). Other ANPs used a benchmark holdout.

Test: Brier minus beta-binomial (paired, 95% CI)

algotune, cmmlu, editbench: no model trained on or was selected with these. 45 tasks x 10 order seeds.



* random-subject split: trained on ai2d_test, bird_sql (validation) and editbench (test). Other ANPs used a benchmark holdout.

Validation: Brier ALC by model and acquisition policy

ai2d_test, bird_sql, financebench. ANP checkpoints were selected here, which flatters them. 82 tasks x 3 seeds.

Macro-task Brier ALC, lower is better. Bold: lowest among rows that never trained on these benchmarks.

model	random	uncertainty	conf. extremes	Bernoulli var.	exp. Brier red.
Constant (train mean)	0.2525	0.2525	0.2525	0.2525	0.2525
Empirical mean	0.1892	0.1906	0.1944	0.1883	0.1921
Beta-binomial	0.1717	0.1726	0.1747	0.1704	0.1731
ANP w64	0.1735	0.1740	0.1759	0.1768	0.1749
ANP w64 dense-prefix	0.1735	0.1748	0.1718	0.1729	0.1719
ANP w64 SAM	**0.1667**	**0.1679**	**0.1677**	**0.1684**	**0.1708**
ANP w64, random split*	0.1519	0.1538	0.1536	0.1540	0.1611
ANP w128, random split*	0.1485	0.1503	0.1521	0.1528	0.1511

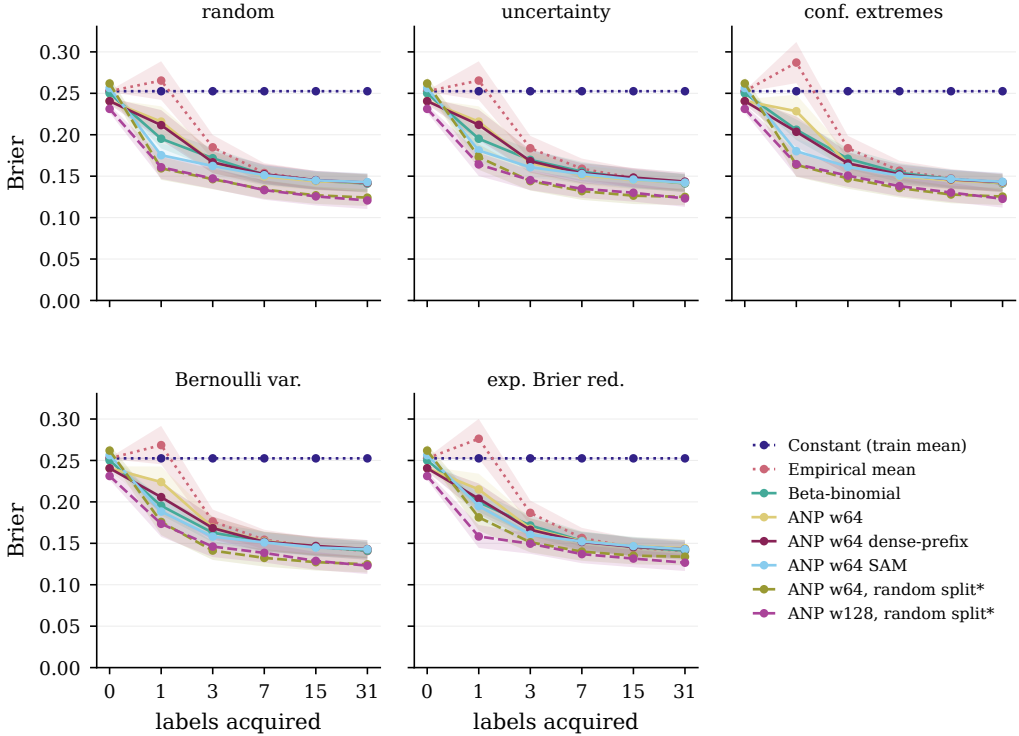
ALC minus beta-binomial under the same policy, paired over tasks: mean ± 95% CI. Bold: CI excludes 0.

model	random	uncertainty	conf. extremes	Bernoulli var.	exp. Brier red.
Constant (train mean)	**+0.0808 ±0.0211**	**+0.0799 ±0.0213**	**+0.0779 ±0.0214**	**+0.0822 ±0.0210**	**+0.0795 ±0.0212**
Empirical mean	**+0.0175 ±0.0056**	**+0.0180 ±0.0056**	**+0.0197 ±0.0058**	**+0.0180 ±0.0069**	**+0.0191 ±0.0088**
ANP w64	+0.0017 ±0.0034	+0.0014 ±0.0034	+0.0012 ±0.0043	**+0.0064 ±0.0060**	+0.0018 ±0.0064
ANP w64 dense-prefix	+0.0017 ±0.0044	+0.0022 ±0.0049	-0.0029 ±0.0067	+0.0026 ±0.0069	-0.0012 ±0.0066
ANP w64 SAM	**-0.0051 ±0.0039**	**-0.0047 ±0.0043**	**-0.0070 ±0.0050**	-0.0019 ±0.0057	-0.0023 ±0.0054
ANP w64, random split*	**-0.0198 ±0.0113**	**-0.0188 ±0.0115**	**-0.0211 ±0.0119**	**-0.0164 ±0.0119**	**-0.0120 ±0.0104**
ANP w128, random split*	**-0.0232 ±0.0110**	**-0.0223 ±0.0107**	**-0.0226 ±0.0115**	**-0.0176 ±0.0128**	**-0.0220 ±0.0093**

* random-subject split: trained on ai2d_test, bird_sql (validation) and editbench (test). Other ANPs used a benchmark holdout.

Validation: Brier by labels acquired (mean ± 1 SE)

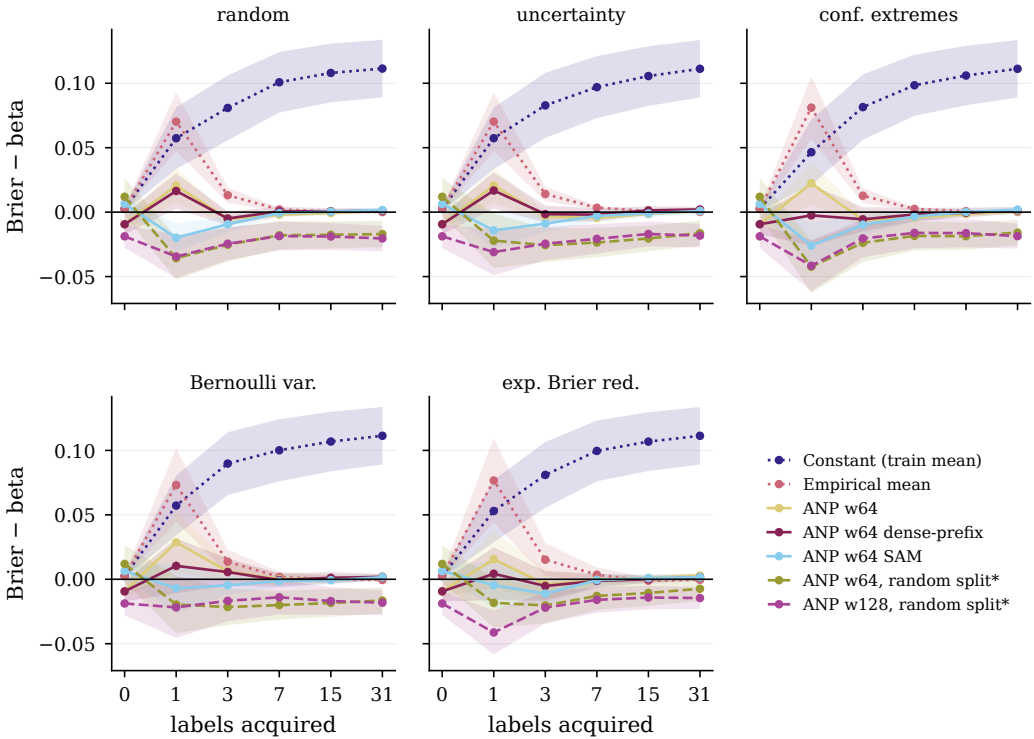
ai2d_test, bird_sql, financebench. ANP checkpoints were selected here, which flatters them. 82 tasks x 3 seeds.



* random-subject split: trained on ai2d_test, bird_sql (validation) and editbench (test). Other ANPs used a benchmark holdout.

Validation: Brier minus beta-binomial (paired, 95% CI)

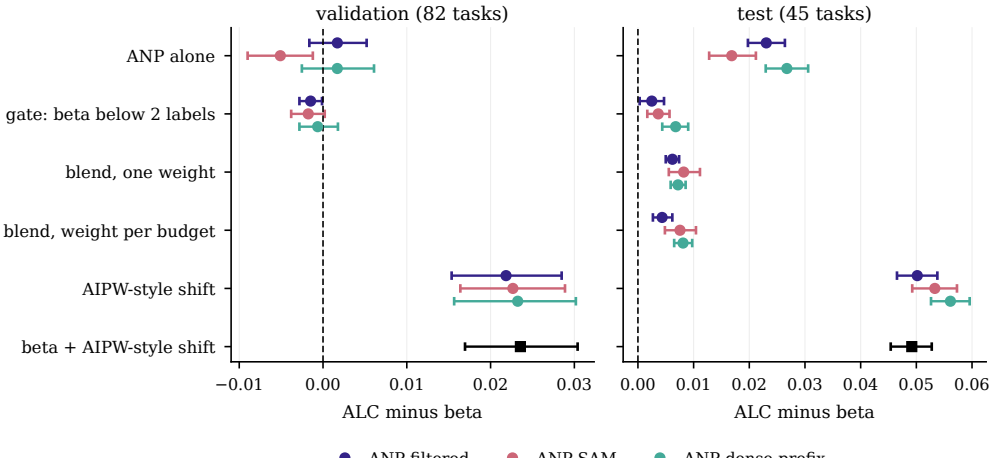
ai2d_test, bird_sql, financebench. ANP checkpoints were selected here, which flatters them. 82 tasks x 3 seeds.



* random-subject split: trained on ai2d_test, bird_sql (validation) and editbench (test). Other ANPs used a benchmark holdout.

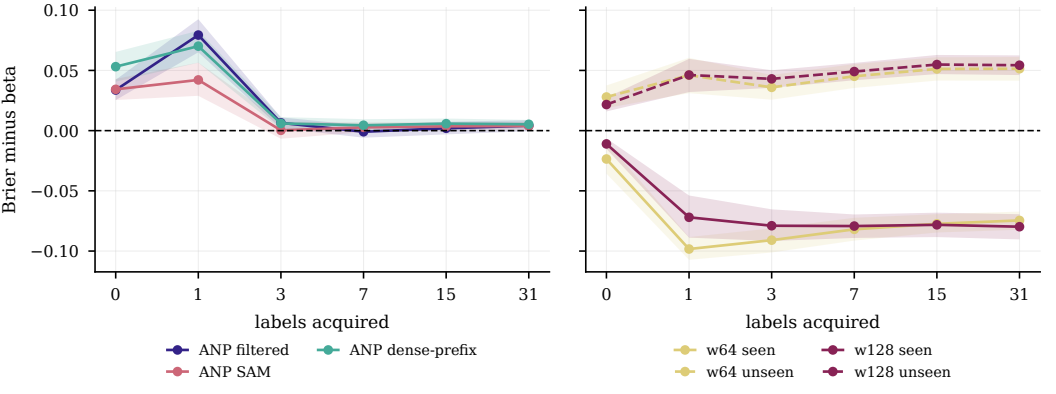
Can beta-binomial rescue the ANP?

Macro Brier-ALC minus beta-binomial, random acquisition; bars: 95% task bootstrap.
Negative = better than beta. Blend weights fitted on validation, so they appear on test only.



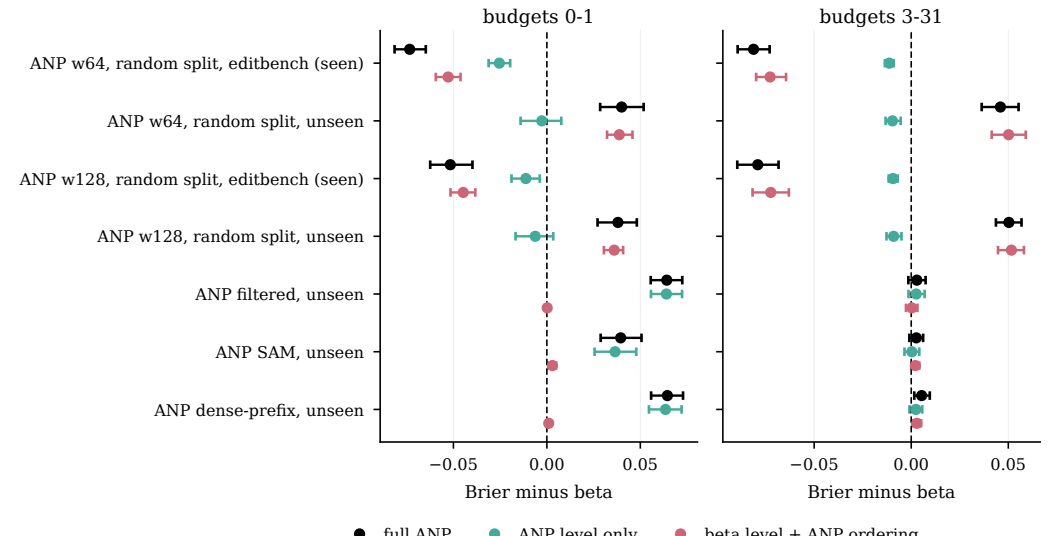
Where the ANP loses: Brier minus beta at each budget (test)

Left: benchmark-holdout ANPs on the three unseen test benchmarks. Right: random-split ANPs on editbench, which they trained on (solid), and on algotune + cmmlu, which they did not (dashed).



Which part fails on a new benchmark: level or item ordering? (test)

Level: the mean of the ANP's predictions for the task. Ordering: each item's offset from that mean.
Brier minus beta; negative = better than beta. Bars: 95% task bootstrap.



Does the ANP's loss grow with subject ability?

One point per task. Ability: the subject's accuracy on the task's acquisition pool (never scored).
Shaded: 5-95% of training-task accuracy (0.00-0.87). No test or validation subject appears in training.

